JAMA Dermatology | Original Investigation

Skin Cancer Diagnosis by Lesion, Physician, and Examination Type
A Systematic Review and Meta-Analysis

Jennifer Y. Chen, MD; Kristen Fernandez, BS; Raj P. Fadadu, MD, MS; Rasika Reddy, MD; Mi-Ok Kim, PhD; Josephine Tan, MLIS; Maria L. Wei, MD, PhD

IMPORTANCE Skin cancer is the most common cancer in the US; accurate detection can minimize morbidity and mortality.

OBJECTIVE To assess the accuracy of skin cancer diagnosis by lesion type, physician specialty and experience, and physical examination method.

DATA SOURCES PubMed, Embase, and Web of Science.

STUDY SELECTION Cross-sectional and case-control studies, randomized clinical trials, and nonrandomized controlled trials that used dermatologists or primary care physicians (PCPs) to examine keratinocytic and/or melanocytic skin lesions were included.

DATA EXTRACTION AND SYNTHESIS Search terms, study objectives, and protocol methods were defined before study initiation. Data extraction was performed by a reviewer, with verification by a second reviewer. A mixed-effects model was used in the data analysis. Data analyses were performed from May 2022 to December 2023.

MAIN OUTCOMES AND MEASURES Meta-analysis of diagnostic accuracy comprised sensitivity and specificity by physician type (primary care physician or dermatologist; experienced or inexperienced) and examination method (in-person clinical examination and/or clinical images vs dermoscopy and/or dermoscopic images).

RESULTS In all, 100 studies were included in the analysis. With experienced dermatologists using clinical examination and clinical images, the sensitivity and specificity for diagnosing keratinocytic carcinomas were 79.0% and 89.1%, respectively; using dermoscopy and dermoscopic images, sensitivity and specificity were 83.7% and 87.4%, and for PCPs, 81.4% and 80.1%. Experienced dermatologists had 2.5-fold higher odds of accurate diagnosis of keratinocytic carcinomas using in-person dermoscopy and dermoscopic images compared with in-person clinical examination and images. When examining for melanoma using clinical examination and images, sensitivity and specificity were 76.9% and 89.1% for experienced dermatologists, 78.3% and 66.2% for inexperienced dermatologists, and 37.5% and 84.6% for PCPs, respectively; whereas when using dermoscopy and dermoscopic images, sensitivity and specificity were 85.7% and 81.3%, 78.0% and 69.5%, and 49.5% and 91.3%, respectively. Experienced dermatologists had 5.7-fold higher odds of accurate diagnosis of melanoma using dermoscopy compared with clinical examination. Compared with PCPs, experienced dermatologists had 13.3-fold higher odds of accurate diagnosis of melanoma using dermoscopic images.

CONCLUSIONS AND RELEVANCE The findings of this systematic review and meta-analysis indicate that there are significant differences in diagnostic accuracy for skin cancer when comparing physician specialty and experience, and examination methods. These summary metrics of clinician diagnostic accuracy could be useful benchmarks for clinical trials, practitioner training, and the performance of emerging technologies.

JAMA Dermatol. 2025;161(2):135-146. doi:10.1001/jamadermatol.2024.4382
Published online November 13, 2024.

Supplemental content

Author Affiliations: San Francisco Veterans Affairs Health Care System, San Francisco, California (Chen, Fernandez, Fadadu, Reddy, Wei); Department of Epidemiology and Biostatistics, University of California, San Francisco (Kim); Helen Diller Family Comprehensive Cancer Center, University of California, San Francisco (Kim, Wei); San Francisco Library, University of California, San Francisco (Tan); Department of Dermatology, University of California, San Francisco (Wei).

Corresponding Author: Maria L. Wei, MD, PhD, University of California San Francisco, 1701 Divisadero St, 3rd Floor, San Francisco, CA 94115.

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Skin cancer is the most frequently diagnosed cancer in the US. Approximately 100 640 new cases of invasive melanoma were diagnosed in 2024[1]; the most recent study of nonmelanoma skin cancer reported 5.4 million cases diagnosed in 2012.[2] The 5-year survival among patients with localized melanoma is greater than 99%, whereas 5-year survival for those with distant or metastatic melanoma falls to 35%.[3] Similarly, 5-year survival among patients with localized squamous cell carcinoma (SCC) is 95%, whereas 5-year survival for those with regional SCC decreases to 58%.[4] Accurate screening for skin cancer coupled with earlier diagnosis can optimize outcomes, minimize the number of invasive diagnostic procedures for benign lesions (eg, skin biopsy), avoid associated morbidities, and reduce health care costs.

Using the naked eye is the current standard of care for skin cancer examination, and histopathologic testing remains the gold standard for skin cancer diagnosis.[5,6] Dermoscopy is a noninvasive in vivo magnification method that identifies features in skin lesions that may not be visible to the naked eye.[7-9] Adding dermoscopy to conventional naked-eye clinical examination in melanoma screening has been associated with more than 40% fewer biopsy procedures performed by dermatologists.[8] Although dermoscopy has been adopted in dermatology training programs in the US, a lack of training has been reported to be a barrier to routine use by primary care physicians (PCPs).[10]

Accurate examination for skin cancer depends on the sensitivity and specificity of the physical examination method. Several large studies have shown an unclear benefit of screening the general adult population; however, the examination sensitivity and specificity of the physicians' specialties were unclear because the accuracy of several practitioner types were reported in aggregate.[11-13] New technologies have been developed for skin cancer examination, including artificial intelligence analysis of images, tape sampling to collect skin surface messenger RNA, reflectance confocal microscopy, and impedance spectroscopy, among others.[14-17] However, there are no standard performance benchmarks with which to compare these technologies. Often, the validation studies include only a small number of dermatologists or PCPs for comparison.[18-21] The emergence of these novel diagnostic tools raises the question of how to best benchmark them against the current standard of care. In 2015, the US National Academies issued a consensus report, *Improving Diagnosis in Health Care*,[22] which focused on supporting efforts toward implementing diagnostic excellence. Taken together, there is a need to assess the current state of diagnostic performance for skin lesion examination and for summary metrics of the diagnostic accuracy of skin cancer detection in both clinical and research settings.

Many previous studies have assessed the diagnostic accuracy of dermatologists and/or PCPs; however, there is considerable heterogeneity in study methodologies: varying ground truth, differing practitioner types, lack of or differing reported statistics. Previous systematic reviews have evaluated the diagnostic accuracy of either exclusively dermatologists, PCPs, or mixed practitioners (not by specialty) or have not stratified studies by practitioners' experience level.[23-28] In addition, most reviews have focused on melanoma or other specific classes of skin cancer.[23,27-33] We conducted a systematic review and meta-analysis to assess the diagnostic accuracy of skin cancer diagnosis by lesion type (keratinocytic or melanocytic), practitioner type (dermatologist or PCP) and dermatologists' experience level, and examination modality (magnified or nonmagnified).

Key Points

Question What is the accuracy of skin cancer diagnosis, stratified by lesion type, physician specialty and experience, and examination method?

Findings This systematic review and meta-analysis including 100 studies found that using dermoscopy compared with clinical examination substantially improved diagnostic accuracy for melanoma (relative odds ratio [ROR], 5.7) and for keratinocytic cancer (ROR, 2.5). Sensitivity and specificity using clinical examination and images of melanoma were 76.9% and 89.1% for experienced dermatologists compared with 78.3% and 66.2% for inexperienced dermatologists and 37.5% and 84.6% for primary care physicians; using in-person dermoscopy and dermoscopic images, they were 85.7% and 81.3% for experienced dermatologists, 78.0% and 69.5% for inexperienced dermatologists, and 49.5% and 91.3% for PCPs.

Meaning These findings indicate that although accuracy varies by lesion type, physician specialty and experience, and examination method, dermatologists had the highest area under the receiver operating characteristic curve using a dermatoscope, and 13.3-fold increased accuracy over primary care physicians when diagnosing melanoma; that said, the data suggest that the greatest accuracy for dermatologists was achieved by using dermoscopy together with clinical examination.

Methods

The study protocol was prospectively registered with PROSPERO (CRD42022283603). The review and analysis were performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) reporting guideline.[34]

Data Sources and Search Strategy

PubMed, Embase, and Web of Science were searched for relevant articles, from database inception to December 31, 2021. The search used the following terms: diagnostic accuracy, dermoscopy, naked eye exam, skin cancer, dermatologist, PCPs, sensitivity, and specificity (**Figure 1**; eTable 1 in Supplement 1).

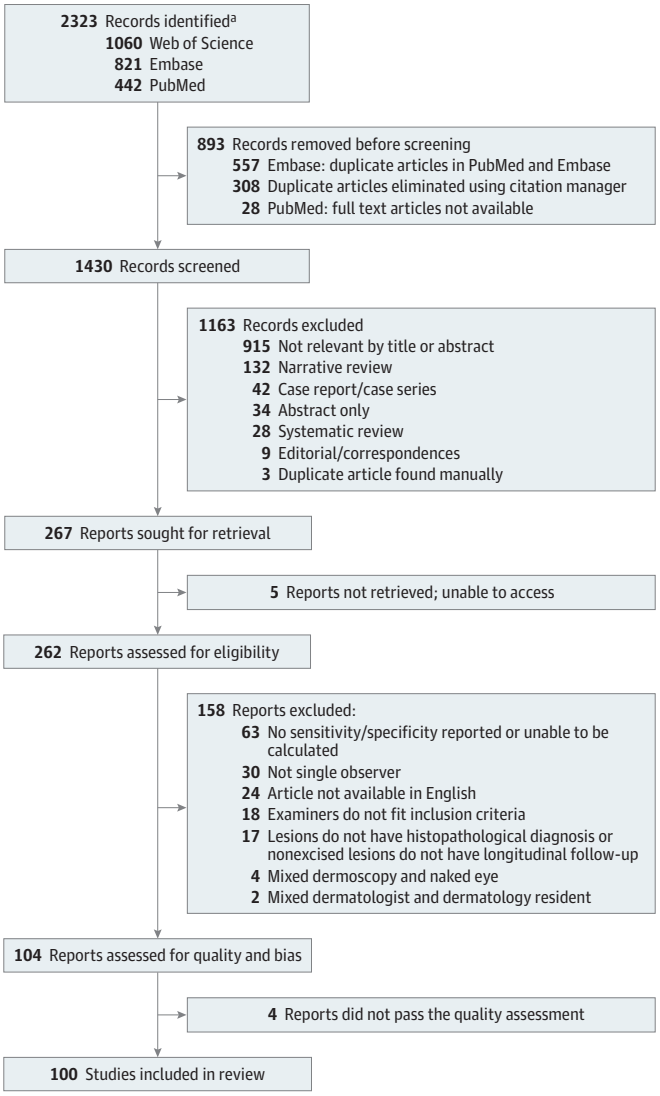
Study Selection

Inclusion and Exclusion Criteria

Study types included were cross-sectional studies, case-control studies, randomized clinical trials, and nonrandomized controlled trials with examination performed by dermatologists or PCPs; those with midlevel practitioners, nurses, medical students, or mixed practitioners as skin examiners were excluded.

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses Flow Diagram



[a]Full search strategy for each database can be found in eTable 1 in Supplement 1.

Target conditions were melanocytic and keratinocytic lesions, including benign lesions and lesions concerning for cutaneous melanoma, SCC, or basal cell carcinoma. Studies of rarer forms of skin cancer were excluded.

Studies that assessed diagnostic accuracy using nonmagnified (eg, unassisted in-person clinical examination, clinical digital images) or magnified methods (eg, in-person dermoscopy, dermoscopic digital images) and reported quantitative measures (eg, sensitivity, specificity, area under the receiver operating characteristic curve [AUROC]) were included. We also included all algorithms or checklists that were used to assist in diagnosis. Acceptable reference standards were histopathologic diagnoses or longitudinal follow-up when lesions were initially designated as benign.

Studies that reported the accuracy of specific morphologic features but not diagnosis, or reported the accuracy of clinical diagnoses but not whether inspection was by naked eye or dermoscopy were excluded. We excluded studies that obtained diagnoses by methods other than single observer (eg, by consensus or averages) because they did not reflect the usual clinical setting, did not report sensitivity and specificity, and/or were not published in English.

Selection Process

Each study was independently screened by 2 of 4 reviewers (J.C., K.F., R.F., R.R.). One of 4 reviewers extracted data, which was then verified by the second reviewer. Two of the 4 reviewers also independently assessed the risk of bias and applicability of included studies using the SIGN (Scottish Intercollegiate Guidelines Network) checklist[35] for diagnostic studies (eTable 3 in Supplement 1). Disagreements were resolved by consensus among 3 reviewers (J.C., K.F., R.R.); an attending dermatologist (M.W.) was available for consultation when disagreements were difficult to resolve. Complete details are available in eTables 2 and 3 in Supplement 1.

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

We initially identified 1430 unique references. After titles and abstracts were screened, 262 full-text articles were reviewed for eligibility. Of these, 104 studies were assessed for quality and bias using the Quality Assessment of Diagnostic Accuracy Studies, Version 2. We excluded 4 studies that did not pass the quality assessment (Figure 1).

Analytic Definitions

This review compared magnified with unmagnified methods of skin lesion examination, and the performance of experienced dermatologists (cutoff range, ≥3-6 years of training/dermoscopy experience), inexperienced dermatologists (cutoff range, <3-6 years of training/dermoscopy experience), and PCPs using those methods. Unmagnified methods were either in-person unassisted (naked eye) clinical examination or used clinical images. Magnified methods were either in-person dermoscopic evaluation or dermoscopic images. Dermoscopic evaluations used a dermoscope, a handheld device that provides magnification and enhanced visualization of skin lesions. Unless otherwise noted, combined unmagnified methods (in-person clinical examination and clinical images; hereafter, clinical examination/images) were compared with combined magnified methods (in-person dermoscopy and dermoscopic images (hereafter, demoscropy/images).

Statistical Analysis

Individual study results were combined for meta-analysis of sensitivity, specificity, AUROC, and relative odds ratio (ROR) of diagnostic accuracy results. For studies that reported head-to-head direct comparisons of examination methods (dermoscopy/images compared with clinical examination/images), the metrics of each examination method (sensitivity and specificity) was summarized as the odds ratio for accurate diagnosis. The relative odds ratios (RORs) for diagnostic accuracy were then computed for comparison within each study first and combined across the studies in a meta-analysis on a log-transformed scale.[36] The standard error of each ROR was defined as the square root of the sum of the odds of each examination method on the log-transformed scale. Two studies had zero false negatives, and we applied the standard adjustment method by adding 0.5 to the zero cell counts. I^2 statistic was computed to quantify the degree of between-study heterogeneity (as the percentage of variation across studies due to between-study heterogeneity).[37] When no head-to-head within study comparisons were available, we used other metrics (ie, sensitivity, specificity, AUROC) to summarize clinician diagnostic accuracy. These summary statistics, based on all eligible studies, gave a more accurate representation of overall sensitivity and specificity. Given that the sensitivity and specificity of a diagnostic test are typically (inversely) associated due to their dependence on each other through a cutoff value, a bivariate diagnostic mixed-effects model was used to account for the dependency between the sensitivity and specificity of a diagnostic test and variability in either statistic between studies.[38,39] Plots and tables show the bivariate diagnostic random-effects meta-analyses separately conducted for each examination method. Lastly, prevalence of melanoma or keratinocyte cancer in the sample may affect diagnostic accuracy. We used meta-regression to identify possible associations. Analysis was conducted in R, version 4.2 (The R Foundation for Statistical Computing) using the mada and metafor packages.[40] Statistical inference was performed using 2-tailed tests, with $P < .05$ considered statistically significant. Data analyses were performed from May 2022 to December 2023.

Results

The meta-analysis included 100 studies (eTable 3 in Supplement 1).[41-140] These studies were conducted in Asia, Europe (the majority), North America, Oceania, and South America, and were published between 1988 and 2021 (eTable 4 in Supplement 1). Only 4 studies reported participants' race and ethnicity in the datasets, and among these, most lesions involved White participants (eTable 5 in Supplement 1). Five studies reported Fitzpatrick skin type in their datasets, mostly types I to III (eTable 5 in Supplement 1).

Keratinocytic Carcinoma

When including all eligible studies that reported accuracy for diagnosis of keratinocytic carcinomas, experienced dermatologists had summary sensitivity of 79.0% (95% CI, 62.8%-89.3%) and specificity of 89.1% (95% CI, 70.5%-96.5%) when assessing lesions using clinical examination/images, and sensitivity of 83.7% (95% CI, 76.6%-89.0%) and specificity of 87.4% (95% CI, 78.9%-92.8%) when using dermoscopy/images (**Table 1**; **Figure 2**A).[41, 44, 45, 65, 73, 77, 79, 81, 85, 90, 94, 95, 112, 113, 131, 133, 135, 139] AUROC was 0.89 when assessed with clinical examination/images, and 0.91 with dermoscopy/images (Table 1; eFigure 1 in Supplement 1). Only 1 study[83] of inexperienced dermatologists diagnosing keratinocytic carcinomas met our eligibility criteria. It reported that dermatologists-in-training using clinical images achieved mean (SD) sensitivity and specificity of 94.2% (5.2%) and 68.5% (11.6%), respectively, with an AUROC of 0.88 (Table 1). PCPs' accuracy for diagnosing keratinocytic carcinomas using dermoscopy had sensitivity and specificity of 81.4% (95% CI, 43.3%-96.1%) and 80.1% (95% CI, 54.2%-93.2%), respectively, with an AUROC of 0.87 (Table 1; Figure 2A).[108,137] No studies of PCPs using clinical examination to diagnose keratinocytic carcinomas met our inclusion criteria. We did not find prevalence of keratinocytic neoplasms in the study datasets to be significantly correlated with sensitivity (eFigure 4 in Supplement 1).

To minimize between-study heterogeneity, we restricted the analysis to studies that directly compared the accuracy for diagnosis of keratinocytic carcinomas by experienced dermatologists using dermoscopy/images compared with clinical examination/images. We found a 2.5-fold (95% CI, 1.3- to 5.0-fold) increase in the odds of accurate diagnosis of keratinocytic carcinomas using dermoscopy/images compared to using clinical examination/images (**Figure 3**A).[44,77,81,90,131,133]

Melanoma

When including all eligible studies, experienced dermatologists had sensitivity of 76.9% (95% CI, 69.3%-83.1%) and speci-

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Table 1. Diagnostic Accuracy for Keratinocytic Carcinomas, Stratified by Physician Type and Experience and Examination Method

Physician	Method	% (95% CI)		AUROC	Studies, No.[a]	Physicians, No.[b]	Lesions, No.
		Sensitivity	Specificity				
Experienced dermatologist	Clinical examination/images	79.0 (62.8-89.3)	89.1 (70.5-96.5)	0.89	9	>16	8441
	Dermoscopy/images	83.7 (76.6-89.0)	87.4 (78.9-92.8)	0.91	18	>303	9564
Primary care physician	Dermoscopy	81.4(43.3-96.1)	80.1 (54.2-93.2)	0.87	2	>50	8515
Inexperienced dermatologist, % (SD)[c]	Clinical images	94.2 (5.2)	68.5 (11.6)	0.88	1	34	80

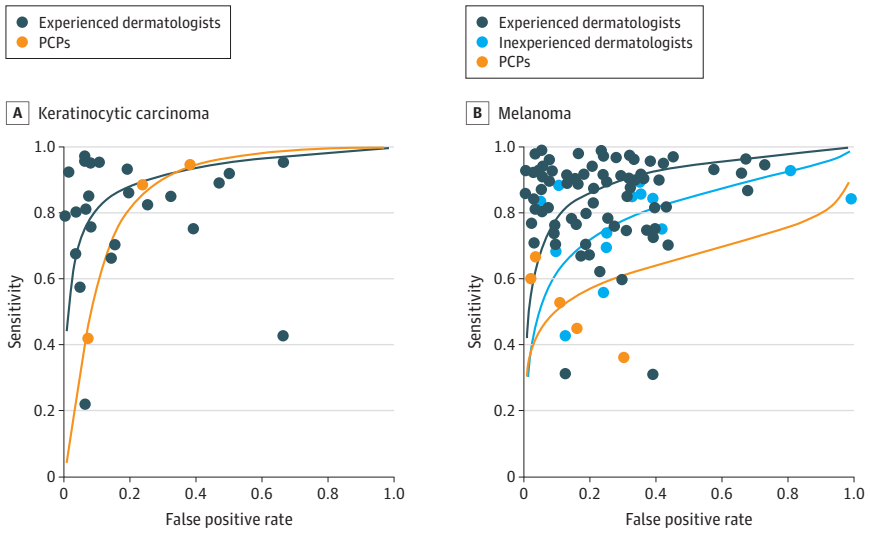
Abbreviation: AUROC, area under the receiver operating characteristic curve.

[a] Some studies were included in multiple physician categories.

[b] Not all studies specified how many physicians participated.

[c] Based on data from only 1 study[83] that met inclusion criteria and reported clinical diagnostic accuracy for inexperienced dermatologists evaluating keratinocytic carcinoma. No studies that met inclusion criteria were found for inexperienced dermatologists using dermoscopy or for PCPs using clinical examination.

Figure 2. Diagnostic Accuracy of Experienced Dermatologists, Inexperienced Dermatologists, and Primary Care Physicians (PCPs) Using Dermoscopy/Images



A, Analysis assessed 27 records from 20 studies that reported results for dermoscopy/images by experienced dermatologists[41,44,45,65,73,77,79,81,85,90,94,95,112,113,131,133,135,139] or PCPs.[108,137] AUROC was 0.91 for experienced dermatologists and 0.87 for PCPs. B, Analysis assessed 92 records from 68 studies that reported results for dermoscopy/images by experienced dermatologists,[41-43,46-55,57,58,60,64,66-68,71,72,74,76,79,80,82,86-102,105-107,110,111,114,116,117,119,121,123,124,126,127,129,131,132,134,138,140] inexperienced dermatologists,[51-54,57,59,82,89,101,116,119,126,127,140] and PCPs.[104,108,114,137,138] AUROC was 0.90 for experienced dermatologists, 0.81 for inexperienced dermatologists, and 0.65 for PCPs. AUROC indicates area under the receiver operating characteristic curve.

ficity of 89.1%, (95% CI, 76.9%-95.3%), with an AUROC of 0.84, when using clinical examination/images, and sensitivity of 85.7% (95% CI, 82.5%-88.3%) and specificity of 81.3% (95% CI, 76.3%-85.4%), respectively, with an AUROC of 0.90, when using dermoscopy/images (**Table 2**; Figure 2B).[41-43,46-55,57,58,60,64,66-68,71,72,74,76,79,80,82,86-102,105-107,110,111,114,116,117,119,121,123,124,126,127,129,131,132,134,138,140] Inexperienced dermatologists diagnosing melanoma had sensitivity of 78.3% (95% CI, 54.9%-91.4%) and specificity of 66.2% (95% CI, 55.9%-75.1%), with an AUROC of 0.73, when assessing clinical images, and sensitivity of 78.0% (95% CI, 69.3%-84.7%) and specificity of 69.5% (95% CI, 52.9%-82.2%), with an AUROC of 0.81, when assessing dermoscopic images (Table 2, Figure 2B).[51-54,57,59,82,89,101,116,119,126,127,140] We did not find eligible studies that reported the accuracy of inexperienced dermatologists diagnosing melanomas in person. PCPs' had sensitivity of 49.5% (95% CI, 40.4%-58.6%) and specificity of 91.3% (95% CI, 78.0%-96.9%), with an AUROC of 0.65, for diagnosis of melanoma using dermoscopy/images (Table 2, Figure 2B).[104,108,114,137,138] The single study[104] of PCPs using clinical examination to diagnose melanoma reported sensitivity of 37.5% (95% CI, 21.1%-56.3%) and specificity of 84.6% (95% CI, 80.0%-88.5%) (Table 2). We did not find prevalence of melanomas in the study datasets to be significantly correlated with sensitivity (eFigure 4 in Supplement 1).

When restricting the analysis to studies that directly compared the use of dermoscopy/images with clinical examination/images by experienced dermatologists, we found that dermatologists had 2.6-fold (95% CI, 0.9- to 7.9-fold) higher odds of accurate diagnosis of melanoma with dermoscopy/images compared with clinical examination/images (Figure 3B).[54,60,68,88,90,110,119,131] When analyses were further restricted to studies that directly compared dermoscopy to clinical examination, dermatologists had 5.7-fold (95% CI, 2.2- to 15.2-fold) higher odds of accurate diagnosis of melanoma using dermoscopy (Figure 3C).[60,68,88,110,131]

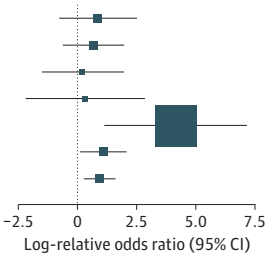
Studies that directly compared the accuracy of experienced dermatologists and PCPs showed that experienced dermatologists had 13.3-fold (95% CI, 7.2- to 24.5-fold) higher odds

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Figure 3. Comparative Diagnostic Accuracy, By Lesion Type, Examination Method (Dermoscopy/Images vs Clinical Examination/Images), and Physician Type (Dermatologist vs Primary Care Physician [PCP])

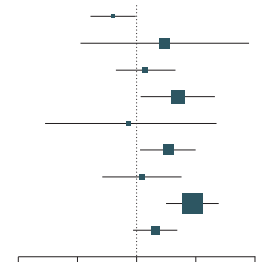
A Experienced dermatologists diagnosing keratinocytic carcinoma

Study	No. of lesions	No. of dermatologists	Magnified sensitivity	Magnified specificity	Nonmagnified sensitivity	Nonmagnified specificity	Log-relative odds ratio (95% CI)
Altameemi et al,[44] 2020	466	NS	0.92	0.50	0.80	0.55	2.4 (0.5-12.1)
Gómez-Martín et al,[77] 2019	228	1	0.85	0.68	0.69	0.74	1.9 (0.5-7.2)
Hacioglu et al,[81] 2013	160	2	0.86	0.80	0.79	0.84	1.2 (0.2-7.0)
Kroemer et al,[90] 2011	208	1	0.97	0.94	0.97	0.91	1.4 (0.1-17.3)
Stanganelli et al,[131] 2000	6744	NS	0.79	1.00	0.49	0.99	63.5 (3.1-1298.3)
Ulrich et al,[133] 2015	470	12	0.89	0.53	0.89	0.23	3.0 (1.1-7.9)
Summary	8276	>16	NA	NA	NA	NA	2.5 (1.3-5.0)



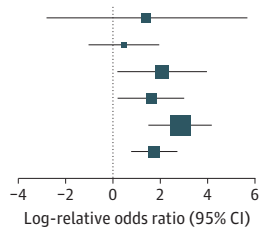
B Experienced dermatologists diagnosing melanoma

Study	No. of lesions	No. of dermatologists	Magnified sensitivity	Magnified specificity	Nonmagnified sensitivity	Nonmagnified specificity	Log-relative odds ratio (95% CI)
Brinker et al,[54] 2019	600	63	0.75	0.64	0.92	0.61	0.3 (0.1-1.0)
Carli et al,[60] 2003	5084	2	0.86	1.00	0.86	0.98	4.2 (0.1-303.0)
Cristofollini et al,[68] 1994	440	4	0.88	0.79	0.85	0.75	1.6 (0.4-7.1)
Krähn et al,[88] 1998	160	1	0.90	0.93	0.79	0.78	8.0 (1.2-53.7)
Kroemer et al,[90] 2011	208	1	0.92	0.97	0.92	0.98	0.7 (0.01-58.0)
Nachbar et al,[110] 1994	344	NS	0.93	0.91	0.84	0.83	5.0 (1.2-20.5)
Rao et al,[119] 1997	144	2	0.91	0.69	0.81	0.78	1.3 (0.2-10.3)
Stanganelli et al,[131] 2000	6744	NS	0.93	1.00	0.67	0.99	17.2 (4.5-65.5)
Summary	13724	>73	NA	NA	NA	NA	2.6 (0.9-7.9)



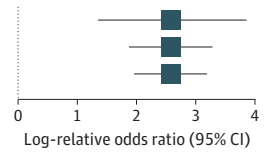
C Experienced dermatologists diagnosing melanomas, in-person only

Study	No. of lesions	No. of dermatologists	Magnified sensitivity	Magnified specificity	Nonmagnified sensitivity	Nonmagnified specificity	Log-relative odds ratio (95% CI)
Carli et al,[60] 2003	5084	2	0.86	1.00	0.86	0.98	4.2 (0.1-303.0)
Cristofollini et al,[68] 1994	440	4	0.88	0.79	0.85	0.75	1.6 (0.4-7.1)
Krähn et al,[88] 1998	160	1	0.90	0.93	0.79	0.78	8.0 (1.2-53.7)
Nachbar et al,[110] 1994	344	NS	0.93	0.91	0.84	0.83	5.0 (1.2-20.5)
Stanganelli et al,[131] 2000	6744	NS	0.93	1.00	0.67	0.99	17.2 (4.5-65.5)
Summary	12772	>7	NA	NA	NA	NA	5.7 (2.2-15.2)



D Dermatologists compared with primary care physicians, using dermoscopic images

Study	No. of lesions	No. of dermatologists	No. of PCPs	Dermatologist sensitivity	Dermatologist specificity	PCP sensitivity	PCP specificity	Log-relative odds ratio (95% CI)
Piccolo et al,[114] 2014	330	3	1	0.91	0.64	0.36	0.70	13.3 (3.9-46.1)
Yu et al,[138] 2018	1448	2	2	0.97	0.67	0.45	0.84	13.2 (6.5-26.9)
Summary	1778	5	3	NA	NA	NA	NA	13.3 (7.2-24.5)



A-C, Random effects meta-analysis comparing examination by magnified (in-person dermoscopy or dermoscopic images) vs nonmagnified (in-person clinical examination or clinical images) methods. A, Each study directly compared magnified vs nonmagnified methods.[44,77,81,90,131,133] for diagnosing keratinocytic carcinoma; I^2 = 10.0%. B, Each study directly compared magnified vs nonmagnified methods[54,60,68,88,90,110,119,131] for diagnosing melanoma; I^2 = 66.5%. C, In person, no images; each study directly compared magnified vs nonmagnified methods[60,68,88,110,131] for diagnosing melanoma; I^2 = 37.0%. D, Each study directly compared dermatologists with PCPs[114,138] in diagnosing melanoma using dermoscopic images. Random effects meta-analysis, with use of dermoscopic images; I^2 was not calculated due to low number of included studies. NA indicates not applicable, and NS, not stated.

of accurate diagnosis of melanoma compared to PCPs when using dermoscopic images (Figure 3D).[114,138] No studies directly comparing dermatologists with PCPs using clinical examination/images met eligibility criteria.

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Table 2. Diagnostic Accuracy for Melanoma, Stratified by Physician Type and Experience and Examination Method

Physician	Method	% (95% CI)		AUROC	Studies, No.[a]	Physicians, No.[b]	Lesions, No.
		Sensitivity	Specificity				
Experienced dermatologist	Clinical examination/images	76.9 (69.3-83.1)	89.1 (76.9-95.3)	0.84	11	>96	11 216
	Dermoscopy/images	85.7 (82.5-88.3)	81.3 (76.3-85.4)	0.90	64	>311	42 694
Inexperienced dermatologist	Clinical images	78.3 (54.9-91.4)	66.2 (55.9-75.1)	0.73	3	133	403
	Dermoscopic images	78.0 (69.3-84.7)	69.5 (52.9-82.2)	0.81	14	227	2270
Primary care physician[c]	Dermoscopy/images	49.5 (40.4-58.6)	91.3 (78.0-96.9)	0.65	5	>116	9491
	Clinical examination	37.5 (21.1-56.3)	84.6 (80.0-88.5)	NA	1	63	374

Abbreviations: AUROC, area under the receiver operating characteristic curve; NA, not applicable.

[a] Some studies were included in multiple physician categories.

[b] Not all studies specified how many physicians participated.

[c] Based on data from only 1 study[104] that met inclusion criteria and reported accuracy of primary care physicians diagnosing melanoma using clinical examination.

Discussion

We conducted a systematic review and meta-analysis of studies reporting the diagnostic accuracy of skin cancer diagnosis, stratified by skin lesion type (melanocytic or keratinocytic), physician specialty and experience, and method of examination (dermoscopy/images vs clinical examination/images). Experienced dermatologists had greater accuracy when diagnosing keratinocytic carcinomas using dermoscopy/images compared with clinical examination/images. For the diagnosis of melanoma, experienced dermatologists' increased accuracy using dermoscopy/images compared with clinical examination/images improved even further and reached statistical significance when the analyses were based solely on comparing dermoscopy with clinical examination, consistent with previous reports that teledermoscopy is less accurate compared to in-person diagnosis of melanoma.[141] Dermatologists had 13.3-fold higher odds of accurate diagnosis of melanoma compared to PCPs, a finding with implications for studies assessing the usefulness of skin cancer screening. Finally, for experienced dermatologists, sensitivity was greater with dermoscopy, but specificity was greater with clinical examination, suggesting that a skin examination should optimally include both methods.

Similar to prior studies,[23,33] we found that dermoscopy was superior to clinical skin examination when experienced dermatologists diagnose melanoma in person and keratinocytic carcinomas in person or using images. Diagnosis of melanoma is more difficult compared with diagnosis of keratinocytic carcinomas, likely producing greater accuracy when dermatologists screen for melanomas using dermatoscopes in person, given that in-person examination provides metadata not available with images alone. Our findings also concur with previous reports[30,33] that diagnostic performance for melanomas depended on the level of dermatologist experience. Also similar to previous reports, we found considerable heterogeneity among studies.[23-25,27,29-31,142-146] Sources of heterogeneity included the small numbers of physicians included in the individual studies, differing training or experience levels of skin examiners, and the heterogeneous nature of datasets or lesions. To minimize heterogeneity among compared studies, we performed subanalyses that included only those studies that reported within-study direct comparisons of diagnostic accuracies of dermoscopic examination compared with clinical examination, and, separately, those that directly compared the diagnostic performance of dermatologists and PCPs.

Our meta-analyses yielded summary metrics for skin cancer diagnostic performance that could be useful in several settings. First, they serve as benchmarks for practitioners' skin cancer diagnostic performance and help guide resident training with regards to target proficiency. Together with recommendations for dermoscopy foundational skills for dermatology residents, benchmarks for skin cancer examination could contribute to standardizing dermatology resident training across programs.[147] In addition, training PCPs in dermoscopy could increase diagnostic accuracy; an expert consensus statement has been developed detailing recommended dermoscopy proficiency skills for PCPs.[148]

Second, the findings of this study suggest metrics for benchmarking emerging technologies. Clinical diagnostic tools that bring precision medicine to skin cancer examination and diagnosis are emerging; these tools range from reflectance confocal microscopy and RNA tape–sampling to computer vision models powered by artificial intelligence, among others.[15-18] To be useful, they need to be benchmarked against summary performance metrics, rather than a small group of practitioners examining a study-specific dataset. The stratification of metrics by specialty is useful because developers can compare their technologies to different practitioner benchmarks, depending on the clinical context for which the diagnostic tool is targeted.

Third, our findings highlight the differences in the diagnostic performance of skin cancer detection between specialties. Major studies assessing the effectiveness of skin cancer screening[149] included predominantly nondermatology physicians and few dermatologists, using a combination of predominantly visual examination and less often dermoscopy. Our findings suggest that large studies with PCPs examining skin among the general adult population should be reconsidered; PCPs receive minimal training on skin examination, and are not routinely trained in dermoscopy.[150-152] Sensitivity for mela-

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

noma detection by PCPs using either dermoscopy or clinical examination was less than 50%. However, PCPs conduct many of the skin examinations in the US, and accessibility to specialists can be limited.[152-155] Developing technologies could assist in augmenting PCPs' diagnostic performance, increase effectiveness of examination, and increase equitable access to high-quality skin examinations.[156] Furthermore, as suggested by the workplace screening study by Schneider et al[157] and expert consensus panels,[158,159] a risk-stratified targeted skin-screening approach by dermatologists may be an effective means of detecting skin cancer at a level sufficient to affect outcomes. Future studies are needed to assess both approaches to skin examination, individually and together.

We identified gaps in the literature and future research opportunities. Few studies assessed PCPs' diagnostic accuracy by clinical examination, the standard of care method used by PCPs. Also, few studies reported the demographic characteristics of the patients whose lesions were examined; of those that did, their patients were mostly White and of Fitzpatrick skin types I to III. There is a need to assess the accuracy of skin examination among more diverse skin types. In addition to summary diagnostic performance statistics by our study, there is a need to develop standardized test datasets with clinical and dermoscopic images for benchmarking new technologies and physician diagnostic capabilities. Given the growing use of advanced practice practitioners for skin cancer diagnosis, future studies could assess their summary diagnostic metrics.

Strengths and Limitations

The strengths of this study include the breadth of the included studies (spanning 5 continents), restricting the inclusion criteria to studies that use longitudinal follow-up for benign lesions and/or histopathologic results as ground truth for diagnosis, and stratifying results by lesion type, practitioner type and experience, and examination method.

Limitations of our study include the small number of studies assessing PCPs and the small numbers of physicians in many of the included studies. Some studies provided morphology only without any clinical information, which may affect clinical applicability given that physicians generally have access to both morphology and clinical information. We were not able to conduct meta-analyses comparing in-person examination with using images, due to the low number of studies in each category. Furthermore, in geographic regions where PCPs specialize in skin cancer diagnosis and are proficient in dermoscopy, their performance may be better than reported in the studies; for example a study in Australia reported that 49% of PCPs have dermoscopy training and 61% of skin or pigmented lesion checks used dermoscopy.[160] In comparison, only 9% of PCPs in the US reported using dermoscopy regularly.[161]

Conclusions

This systematic review and meta-analysis of diagnostic accuracy for melanoma and keratinocytic carcinomas by dermatologists and PCPs compared examination modality. We found significant differences in diagnostic accuracy by different physician specialty; accuracy improved when dermoscopic methods were used, and improved further with in-person dermoscopy. These results suggest that clinical examination combined with dermoscopy likely yields better accuracy than either method alone.

ARTICLE INFORMATION

Accepted for Publication: August 27, 2024.

Published Online: November 13, 2024. doi:10.1001/jamadermatol.2024.4382

Author Contributions: Drs Wei and Kim had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Dr Chen and Ms Fernandez contributed equally to this work as co-first authors.
Concept and design: Chen, Fernandez, Fadadu, Reddy, Wei.
Acquisition, analysis, or interpretation of data: All authors.
Drafting of the manuscript: Chen, Fernandez, Fadadu, Kim, Wei.
Critical review of the manuscript for important intellectual content: All authors.
Statistical analysis: Chen, Fadadu, Kim.
Administrative, technical, or material support: Fernandez, Fadadu, Reddy, Wei.
Supervision: Wei.

Conflict of Interest Disclosures: None reported.

Meeting Presentation: This paper was presented at the 2024 Annual Meeting of the Society for Investigative Dermatology; May 16, 2024; Dallas, Texas.

Data Sharing Statement: See Supplement 2.

REFERENCES

1. Siegel RL, Giaquinto AN, Jemal A. Cancer statistics, 2024. *CA Cancer J Clin*. 2024;74(1):12-49. doi:10.3322/caac.21820

2. Rogers HW, Weinstock MA, Feldman SR, Coldiron BM. Incidence estimate of nonmelanoma skin cancer (keratinocyte carcinomas) in the U.S. population, 2012. *JAMA Dermatol*. 2015;151(10):1081-1086. doi:10.1001/jamadermatol.2015.1187

3. American Cancer Society. Survival Rates for Melanoma Skin Cancer. Accessed August 3, 2024. https://www.cancer.org/cancer/types/melanoma-skin-cancer/detection-diagnosis-staging/survival-rates-for-melanoma-skin-cancer-by-stage.html

4. Eisemann N, Jansen L, Castro FA, et al; GEKID Cancer Survival Working Group. Survival with nonmelanoma skin cancer in Germany. *Br J Dermatol*. 2016;174(4):778-785. doi:10.1111/bjd.14352

5. Ulrich M, Stockfleth E, Roewert-Huber J, Astner S. Noninvasive diagnostic tools for nonmelanoma skin cancer. *Br J Dermatol*. 2007;157(suppl 2):56-58. doi:10.1111/j.1365-2133.2007.08275.x

6. Rigel DS, Russak J, Friedman R. The evolution of melanoma diagnosis: 25 years beyond the ABCDs. *CA Cancer J Clin*. 2010;60(5):301-316. doi:10.3322/caac.20074

7. Wolner ZJ, Yélamos O, Liopyris K, Rogers T, Marchetti MA, Marghoob AA. Enhancing skin cancer diagnosis with dermoscopy. *Dermatol Clin*. 2017;35(4):417-437. doi:10.1016/j.det.2017.06.003

8. Carli P, de Giorgi V, Chiarugi A, et al. Addition of dermoscopy to conventional naked-eye examination in melanoma screening: a randomized study. *J Am Acad Dermatol*. 2004;50(5):683-689. doi:10.1016/j.jaad.2003.09.009

9. Argenziano G, Soyer HP. Dermoscopy of pigmented skin lesion: a valuable tool for early diagnosis of melanoma. *Lancet Oncol*. 2001;2(7):443-449. doi:10.1016/S1470-2045(00)00422-8

10. Fee JA, McGrady FP, Hart ND. Dermoscopy use in primary care: a qualitative study with general practitioners. *BMC Prim Care*. 2022;23(1):47. doi:10.1186/s12875-022-01653-7

11. Kaiser M, Schiller J, Schreckenberger C. The effectiveness of a population-based skin cancer screening program: evidence from Germany. *Eur J Health Econ*. 2018;19(3):355-367. doi:10.1007/s10198-017-0888-4

12. Katalinic A, Eisemann N, Waldmann A. Skin cancer screening in Germany. Documenting melanoma incidence and mortality from 2008 to 2013. *Dtsch Arztebl Int*. 2015;112(38):629-634.

13. Datzmann T, Schoffer O, Meier F, Seidler A, Schmitt J. Are patients benefiting from participation in the German skin cancer screening

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

programme? A large cohort study based on administrative data. *Br J Dermatol*. 2022;186(1):69-77. doi:10.1111/bjd.20658

14. Pflugfelder A, Kochs C, Blum A, et al; German Dermatological Society; DermatologicCooperative Oncology Group. Malignant melanoma S3-guideline "diagnosis, therapy and follow-up of melanoma". *J Dtsch Dermatol Ges*. 2013;11(Suppl 6)(suppl 6):1-116, 1-126. doi:10.1111/ddg.12113_suppl

15. Wachsman W, Morhenn V, Palmer T, et al. Noninvasive genomic detection of melanoma. *Br J Dermatol*. 2011;164(4):797-806. doi:10.1111/j.1365-2133.2011.10239.x

16. Brouha B, Ferris LK, Skelsey MK, et al. Real-world utility of a non-invasive gene expression test to rule out primary cutaneous melanoma: a large US registry study. *J Drugs Dermatol*. 2020;19(3):257-262. doi:10.36849/JDD.2020.4766

17. Young AT, Vora NB, Cortez J, et al. The role of technology in melanoma screening and diagnosis. *Pigment Cell Melanoma Res*. 2021;34(2):288-300. doi:10.1111/pcmr.12907

18. Yu Z, Nguyen J, Nguyen TD, et al. Early melanoma diagnosis with sequential dermoscopic images. *IEEE Trans Med Imaging*. 2022;41(3):633-646. doi:10.1109/TMI.2021.3120091

19. Brinker TJ, Hekler A, Enk AH, et al. Deep neural networks are superior to dermatologists in melanoma image classification. *Eur J Cancer*. 2019;119:11-17. doi:10.1016/j.ejca.2019.05.023

20. Esteva A, Kuprel B, Novoa RA, et al. Dermatologist-level classification of skin cancer with deep neural networks. *Nature*. 2017;542(7639):115-118. doi:10.1038/nature21056

21. Krakowski I, Kim J, Cai ZR, et al. Human-AI interaction in skin cancer diagnosis: a systematic review and meta-analysis. *NPJ Digit Med*. 2024;7(1):78. doi:10.1038/s41746-024-01031-w

22. National Academy of Sciences; Committee on Diagnostic Error in Health Care; Board on Health Care Services; Institute of Medicine; the National Academies of Sciences, Engineering, and Medicine. In: Balogh EP, Miller BT, Ball JR, eds. *Improving Diagnosis in Health Care. National Academies Press*. 2015.

23. Dinnes J, Deeks JJ, Chuchu N, et al; Cochrane Skin Cancer Diagnostic Test Accuracy Group. Visual inspection and dermoscopy, alone or in combination, for diagnosing keratinocyte skin cancers in adults. *Cochrane Database Syst Rev*. 2018;12(12):CD011901. doi:10.1002/14651858.CD011901.pub2

24. Mayer J. Systematic review of the diagnostic accuracy of dermatoscopy in detecting malignant melanoma. *Med J Aust*. 1997;167(4):206-210. doi:10.5694/j.1326-5377.1997.tb138847.x

25. Bafounta ML, Beauchet A, Aegerter P, Saiag P. Is dermoscopy (epiluminescence microscopy) useful for the diagnosis of melanoma? Results of a meta-analysis using techniques adapted to the evaluation of diagnostic tests. *Arch Dermatol*. 2001;137(10):1343-1350. doi:10.1001/archderm.137.10.1343

26. Jones OT, Jurascheck LC, van Melle MA, et al. Dermoscopy for melanoma detection and triage in primary care: a systematic review. *BMJ Open*. 2019;9(8):e027529. doi:10.1136/bmjopen-2018-027529

27. Harrington E, Clyne B, Wesseling N, et al. Diagnosing malignant melanoma in ambulatory care: a systematic review of clinical prediction rules. *BMJ Open*. 2017;7(3):e014096. doi:10.1136/bmjopen-2016-014096

28. Vestergaard ME, Macaskill P, Holt PE, Menzies SW. Dermoscopy compared with naked eye examination for the diagnosis of primary melanoma: a meta-analysis of studies performed in a clinical setting. *Br J Dermatol*. 2008;159(3):669-676. doi:10.1111/j.1365-2133.2008.08713.x

29. Reiter O, Mimouni I, Gdalevich M, et al. The diagnostic accuracy of dermoscopy for basal cell carcinoma: a systematic review and meta-analysis. *J Am Acad Dermatol*. 2019;80(5):1380-1388. doi:10.1016/j.jaad.2018.12.026

30. Dinnes J, Deeks JJ, Grainge MJ, et al; Cochrane Skin Cancer Diagnostic Test Accuracy Group. Visual inspection for diagnosing cutaneous melanoma in adults. *Cochrane Database Syst Rev*. 2018;12(12):CD013194. doi:10.1002/14651858.CD013194

31. Nie T, Jiang X, Zheng B, Zhang Y. Effect of reflectance confocal microscopy compared to dermoscopy in the diagnostic accuracy of lentigo maligna: a meta-analysis. *Int J Clin Pract*. 2021;75(8):e14346. doi:10.1111/ijcp.14346

32. Phillips M, Greenhalgh J, Marsden H, Palamaras I. Detection of malignant melanoma using artificial intelligence: an observational study of diagnostic accuracy. *Dermatol Pract Concept*. 2019;10(1):e2020011. doi:10.5826/dpc.1001a11

33. Kittler H, Pehamberger H, Wolff K, Binder M. Diagnostic accuracy of dermoscopy. *Lancet Oncol*. 2002;3(3):159-165. doi:10.1016/S1470-2045(02)00679-4

34. Moher D, Liberati A, Tetzlaff J, Altman DG; PRISMA Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *BMJ*. 2009;339:b2535. doi:10.1136/bmj.b2535

35. Ma LL, Wang YY, Yang ZH, Huang D, Weng H, Zeng XT. Methodological quality (risk of bias) assessment tools for primary and secondary medical studies: what are they and which is better? *Mil Med Res*. 2020;7(1):7. doi:10.1186/s40779-020-00238-8

36. DerSimonian R, Laird N. Meta-analysis in clinical trials. *Control Clin Trials*. 1986;7(3):177-188. doi:10.1016/0197-2456(86)90046-2

37. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Stat Med*. 2002;21(11):1539-1558. doi:10.1002/sim.1186

38. Doebler P, Holling H, Böhning D. A mixed model approach to meta-analysis of diagnostic studies with binary test outcome. *Psychol Methods*. 2012;17(3):418-436. doi:10.1037/a0028091

39. Holling H, Böhning W, Böhning D. Meta-analysis of diagnostic studies based upon SROC-curves: a mixed model approach using the Lehmann family. *Stat Model*. 2012;12(4):347-375. doi:10.1177/1471082X1201200403

40. Viechtbauer W. Conducting meta-analyses in R with the metafor package. *J Stat Softw*. 2010;36(3):1-48. doi:10.18637/jss.v036.i03

41. Ahnlide I, Bjellerup M. Accuracy of clinical skin tumour diagnosis in a dermatological setting. *Acta Derm Venereol*. 2013;93(3):305-308. doi:10.2340/00015555-1560

42. Ahnlide I, Bjellerup M, Nilsson F, Nielsen K. Validity of ABCD rule of dermoscopy in clinical practice. *Acta Derm Venereol*. 2016;96(3):367-372. doi:10.2340/00015555-2239

43. Alarcon I, Carrera C, Palou J, Alos L, Malvehy J, Puig S. Impact of in vivo reflectance confocal microscopy on the number needed to treat melanoma in doubtful lesions. *Br J Dermatol*. 2014;170(4):802-808. doi:10.1111/bjd.12678

44. Altameemi FF. Basal cell carcinoma in Iraq: an observational study. *Med Sci*. 2020;24(106):3918-3922.

45. Altamura D, Menzies SW, Argenziano G, et al. Dermatoscopy of basal cell carcinoma: morphologic variability of global and local features and accuracy of diagnosis. *J Am Acad Dermatol*. 2010;62(1):67-75. doi:10.1016/j.jaad.2009.05.035

46. Antonio JR, Soubhia RM, D'Avila SC, Caldas AC, Trídico LA, Alves FT. Correlation between dermoscopic and histopathological diagnoses of atypical nevi in a dermatology outpatient clinic of the Medical School of São José do Rio Preto, SP, Brazil. *An Bras Dermatol*. 2013;88(2):199-203. doi:10.1590/S0365-05962013000200002

47. Avilés-Izquierdo JA, Ciudad-Blanco C, Sánchez-Herrero A, Mateos-Mayo A, Nieto-Benito LM, Rodríguez-Lomba E. Dermoscopy of cutaneous melanoma metastases: a color-based pattern classification. *J Dermatol*. 2019;46(7):564-569. doi:10.1111/1346-8138.14926

48. Berglund S, Bogren L, Paoli J. Diagnostic accuracy and safety of short-term teledermoscopic monitoring of atypical melanocytic lesions. *J Eur Acad Dermatol Venereol*. 2020;34(6):1233-1239. doi:10.1111/jdv.16144

49. Biyik Ozkaya D, Onsun N, Su O, Arda Ulusal H, Pirmit S. Pitfalls of an automated dermoscopic analysis system in the differential diagnosis of melanocytic lesions. *Acta Dermatovenerol Croat*. 2014;22(4):278-283.

50. Blum A, Luedtke H, Ellwanger U, Schwabe R, Rassner G, Garbe C. Digital image analysis for diagnosis of cutaneous melanoma. Development of a highly effective computer algorithm based on analysis of 837 melanocytic lesions. *Br J Dermatol*. 2004;151(5):1029-1038. doi:10.1111/j.1365-2133.2004.06210.x

51. Blum A, Hofmann-Wellenhof R, Luedtke H, et al. Value of the clinical history for different users of dermoscopy compared with results of digital image analysis. *J Eur Acad Dermatol Venereol*. 2004;18(6):665-669. doi:10.1111/j.1468-3083.2004.01044.x

52. Borsari S, Peccerillo F, Pampena R, et al. The presence of eccentric hyperpigmentation should raise the suspicion of melanoma. *J Eur Acad Dermatol Venereol*. 2020;34(12):2802-2808. doi:10.1111/jdv.16604

53. Brinker TJ, Hekler A, Enk AH, et al; Collaborators. A convolutional neural network trained with dermoscopic images performed on par with 145 dermatologists in a clinical melanoma image classification task. *Eur J Cancer*. 2019;111:148-154. doi:10.1016/j.ejca.2019.02.005

54. Brinker TJ, Hekler A, Hauschild A, et al. Comparing artificial intelligence algorithms to 157 German dermatologists: the melanoma classification benchmark. *Eur J Cancer*. 2019;111:30-37. doi:10.1016/j.ejca.2018.12.016

55. Buhl T, Hansen-Hagge C, Korpas B, et al. Integrating static and dynamic features of melanoma: the DynaMel algorithm. *J Am Acad*

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Dermatol. 2012;66(1):27-36. doi:10.1016/j.jaad.2010.09.731

56. Burroni M, Wollina U, Torricelli R, et al. Impact of digital dermoscopy analysis on the decision to follow up or to excise a pigmented skin lesion: a multicentre study. *Skin Res Technol*. 2011;17(4):451-460. doi:10.1111/j.1600-0846.2011.00518.x

57. Cacciapuoti S, Di Leo G, Ferro M, et al. A measurement software for professional training in early detection of melanoma. *Appl Sci (Basel)*. 2020;10(12):4351. doi:10.3390/app10124351

58. Carli P, De Giorgi V, Giannotti B. Dermoscopy as a second step in the diagnosis of doubtful pigmented skin lesions: how great is the risk of missing a melanoma? *J Eur Acad Dermatol Venereol*. 2001;15(1):24-26. doi:10.1046/j.1468-3083.2001.00147.x

59. Carli P, Quercioli E, Sestini S, et al. Pattern analysis, not simplified algorithms, is the most reliable method for teaching dermoscopy for melanoma diagnosis to residents in dermatology. *Br J Dermatol*. 2003;148(5):981-984. doi:10.1046/j.1365-2133.2003.05023.x

60. Carli P, Mannone F, De Giorgi V, Nardini P, Chiarugi A, Giannotti B. The problem of false-positive diagnosis in melanoma screening: the impact of dermoscopy. *Melanoma Res*. 2003;13(2):179-182. doi:10.1097/00008390-200304000-00011

61. Carrera C, Segura S, Aguilera P, et al. Dermoscopy improves the diagnostic accuracy of melanomas clinically resembling seborrheic keratosis: cross-sectional study of the ability to detect seborrheic keratosis-like melanomas by a group of dermatologists with varying degrees of experience. *Dermatology*. 2017;233(6):471-479. doi:10.1159/000486851

62. Cho SI, Sun S, Mun JH, et al. Dermatologist-level classification of malignant lip diseases using a deep convolutional neural network. *Br J Dermatol*. 2020;182(6):1388-1394. doi:10.1111/bjd.18459

63. Cinotti E, Labeille B, Debarbieux S, et al. Dermoscopy vs. reflectance confocal microscopy for the diagnosis of lentigo maligna. *J Eur Acad Dermatol Venereol*. 2018;32(8):1284-1291. doi:10.1111/jdv.14791

64. Congalton AT, Oakley AM, Rademaker M, Bramley D, Martin RCW. Successful melanoma triage by a virtual lesion clinic (teledermatoscopy). *J Eur Acad Dermatol Venereol*. 2015;29(12):2423-2428. doi:10.1111/jdv.13309

65. Cooper SM, Wojnarowska F. The accuracy of clinical diagnosis of suspected premalignant and malignant skin lesions in renal transplant recipients. *Clin Exp Dermatol*. 2002;27(6):436-438. doi:10.1046/j.1365-2230.2002.01069.x

66. Coras B, Glaessl A, Kinateder J, et al. Teledermatoscopy in daily routine–results of the first 100 cases. *Curr Probl Dermatol*. 2003;32:207-212. doi:10.1159/000067368

67. Costa J, Ortiz-Ibañez K, Salerni G, et al. Dermoscopic patterns of melanoma metastases: interobserver consistency and accuracy for metastasis recognition. *Br J Dermatol*. 2013;169(1):91-99. doi:10.1111/bjd.12314

68. Cristofolini M, Zumiani G, Bauer P, Cristofolini P, Boi S, Micciolo R. Dermatoscopy: usefulness in the differential diagnosis of cutaneous pigmentary lesions. *Melanoma Res*. 1994;4(6):391-394. doi:10.1097/00008390-199412000-00008

69. De Bedout V, Williams NM, Muñoz AM, et al. Skin cancer and dermoscopy training for primary care physicians: a pilot study. *Dermatol Pract Concept*. 2021;11(1):e2021145. doi:10.5826/dpc.1101a145

70. De Giorgi V, Grazzini M, Rossari S, et al. Adding dermatoscopy to naked eye examination of equivocal melanocytic skin lesions: effect on intention to excise by general dermatologists. *Clin Exp Dermatol*. 2011;36(3):255-259. doi:10.1111/j.1365-2230.2010.03963.x

71. di Meo N, Stinco G, Bonin S, et al. CASH algorithm versus 3-point checklist and its modified version in evaluation of melanocytic pigmented skin lesions: The 4-point checklist. *J Dermatol*. 2016;43(6):682-685. doi:10.1111/1346-8138.13201

72. Dreiseitl S, Binder M, Hable K, Kittler H. Computer versus human diagnosis of melanoma: evaluation of the feasibility of an automated diagnostic system in a prospective clinical trial. *Melanoma Res*. 2009;19(3):180-184. doi:10.1097/CMR.0b013e32832a1e41

73. Ermertcan AT, Oztürk F, Gençoğlan G, Eskiizmir G, Temiz P, Horasan GD. Sensitivity, predictive values, pretest-posttest probabilities, and likelihood ratios of presurgery clinical diagnosis of nonmelanoma skin cancers. *Cutan Ocul Toxicol*. 2011;30(1):50-54. doi:10.3109/15569527.2010.521227

74. Feci L, Cevenini G, Nami N, et al. Influence of ambient stressors and time constraints on diagnostic accuracy of borderline pigmented skin lesions. *Dermatology*. 2015;231(3):269-273. doi:10.1159/000435951

75. Frühauf J, Leinweber B, Fink-Puches R, et al. Patient acceptance and diagnostic utility of automated digital image analysis of pigmented skin lesions. *J Eur Acad Dermatol Venereol*. 2012;26(3):368-372. doi:10.1111/j.1468-3083.2011.04081.x

76. Glud M, Gniadecki R, Drzewiecki KT. Spectrophotometric intracutaneous analysis versus dermoscopy for the diagnosis of pigmented skin lesions: prospective, double-blind study in a secondary reference centre. *Melanoma Res*. 2009;19(3):176-179. doi:10.1097/CMR.0b013e328322fe5f

77. Gómez-Martín I, Moreno S, Duran X, Pujol RM, Segura S. Diagnostic accuracy of non-melanocytic pink flat skin lesions on the legs: dermoscopic and reflectance confocal microscopy evaluation. *Acta Derm Venereol*. 2019;99(1):33-40.

78. Green A, Leslie D, Weedon D. Diagnosis of skin cancer in the general population: clinical accuracy in the Nambour survey. *Med J Aust*. 1988;148(9):447-450. doi:10.5694/j.1326-5377.1988.tb139568.x

79. Guitera P, Menzies SW, Argenziano G, et al. Dermoscopy and in vivo confocal microscopy are complementary techniques for diagnosis of difficult amelanotic and light-coloured skin lesions. *Br J Dermatol*. 2016;175(6):1311-1319. doi:10.1111/bjd.14749

80. Guitera P, Pellacani G, Longo C, Seidenari S, Avramidis M, Menzies SW. In vivo reflectance confocal microscopy enhances secondary evaluation of melanocytic lesions. *J Invest Dermatol*. 2009;129(1):131-138. doi:10.1038/jid.2008.193

81. Hacioglu S, Saricaoglu H, Baskan EB, Uner SI, Aydogan K, Tunali S. The value of spectrophotometric intracutaneous analysis in the noninvasive diagnosis of nonmelanoma skin cancers. *Clin Exp Dermatol*. 2013;38(5):464-469. doi:10.1111/j.1365-2230.2012.04460.x

82. Haenssle HA, Fink C, Schneiderbauer R, et al; Reader study level-I and level-II Groups. Man against machine: diagnostic performance of a deep learning convolutional neural network for dermoscopic melanoma recognition in comparison to 58 dermatologists. *Ann Oncol*. 2018;29(8):1836-1842. doi:10.1093/annonc/mdy166

83. Han SS, Moon IJ, Lim W, et al. Keratinocytic skin cancer detection on the face using region-based convolutional neural network. *JAMA Dermatol*. 2020;156(1):29-37. doi:10.1001/jamadermatol.2019.3807

84. Han SS, Moon IJ, Kim SH, et al. Assessment of deep neural networks for the diagnosis of benign and malignant skin neoplasms in comparison with dermatologists: a retrospective validation study. *PLoS Med*. 2020;17(11):e1003381. doi:10.1371/journal.pmed.1003381

85. Hattier GA, Duffy RF, Finkelstein MJ, Beggs SM, Lee JB. Diagnosis and treatment of low-risk superficial basal cell carcinoma in a single visit. *J Dermatolog Treat*. 2022;33(1):191-194. doi:10.1080/09546634.2020.1737637

86. Henning JS, Dusza SW, Wang SQ, et al. The CASH (color, architecture, symmetry, and homogeneity) algorithm for dermoscopy. *J Am Acad Dermatol*. 2007;56(1):45-52. doi:10.1016/j.jaad.2006.09.003

87. Kittler H, Seltenheim M, Pehamberger H, Wolff K, Binder M. Diagnostic informativeness of compressed digital epiluminescence microscopy images of pigmented skin lesions compared with photographs. *Melanoma Res*. 1998;8(3):255-260. doi:10.1097/00008390-199806000-00008

88. Krähn G, Gottlöber P, Sander C, Peter RU. Dermatoscopy and high frequency sonography: two useful non-invasive methods to increase preoperative diagnostic accuracy in pigmented skin lesions. *Pigment Cell Res*. 1998;11(3):151-154. doi:10.1111/j.1600-0749.1998.tb00725.x

89. Kreusch J, Rassner G, Trahn C, Pietsch-Breitfeld B, Henke D, Selbmann HK. Epiluminescent microscopy: a score of morphological features to identify malignant melanoma. *Pigment Cell Res*. 1992;3(S2)(suppl 2):295-298. doi:10.1111/j.1600-0749.1990.tb00388.x

90. Kroemer S, Frühauf J, Campbell TM, et al. Mobile teledermatology for skin tumour screening: diagnostic accuracy of clinical and dermoscopic image tele-evaluation using cellular phones. *Br J Dermatol*. 2011;164(5):973-979. doi:10.1111/j.1365-2133.2011.10208.x

91. Lallas A, Kyrgidis A, Koga H, et al. The BRAAFF checklist: a new dermoscopic algorithm for diagnosing acral melanoma. *Br J Dermatol*. 2015;173(4):1041-1049. doi:10.1111/bjd.14045

92. Langley RGB, Walsh N, Sutherland AE, et al. The diagnostic accuracy of in vivo confocal scanning laser microscopy compared to dermoscopy of benign and malignant melanocytic lesions: a prospective study. *Dermatology*. 2007;215(4):365-372. doi:10.1159/000109087

93. Longo C, Barquet V, Hernandez E, et al. Dermoscopy comparative approach for early diagnosis in familial melanoma: influence of MC1R genotype. *J Eur Acad Dermatol Venereol*. 2021;35(2):403-410. doi:10.1111/jdv.16679

94. Lorentzen HF, Eefsen RL, Weismann K. Comparison of classical dermatoscopy and acrylic

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

globe magnifier dermatoscopy. *Acta Derm Venereol*. 2008;88(2):139-142. doi:10.2340/00015555-0374

95. Łudzik J, Witkowski AM, Roterman-Konieczna I, Bassoli S, Farnetani F, Pellacani G. Improving diagnostic accuracy of dermoscopically equivocal pink cutaneous lesions with reflectance confocal microscopy in telemedicine settings: double reader concordance evaluation of 316 cases. *PLoS One*. 2016;11(9):e0162495. doi:10.1371/journal.pone.0162495

96. MacKie RM, Fleming C, McMahon AD, Jarrett P. The use of the dermatoscope to identify early melanoma using the three-colour test. *Br J Dermatol*. 2002;146(3):481-484. doi:10.1046/j.1365-2133.2002.04587.x

97. MacLellan AN, Price EL, Publicover-Brouwer P, et al. The use of noninvasive imaging techniques in the diagnosis of melanoma: a prospective diagnostic accuracy study. *J Am Acad Dermatol*. 2021;85(2):353-359. doi:10.1016/j.jaad.2020.04.019

98. Maier T, Kulichova D, Schotten K, et al. Accuracy of a smartphone application using fractal image analysis of pigmented moles compared to clinical diagnosis and histological result. *J Eur Acad Dermatol Venereol*. 2015;29(4):663-667. doi:10.1111/jdv.12648

99. Malvehy J, Hauschild A, Curiel-Lewandrowski C, et al. Clinical performance of the Nevisense system in cutaneous melanoma detection: an international, multicentre, prospective and blinded clinical trial on efficacy and safety. *Br J Dermatol*. 2014;171(5):1099-1107. doi:10.1111/bjd.13121

100. Marchetti MA, Codella NCF, Dusza SW, et al; International Skin Imaging Collaboration. Results of the 2016 International Skin Imaging Collaboration International Symposium on Biomedical Imaging challenge: Comparison of the accuracy of computer algorithms to dermatologists for the diagnosis of melanoma from dermoscopic images. *J Am Acad Dermatol*. 2018;78(2):270-277.e1. doi:10.1016/j.jaad.2017.08.016

101. Marchetti MA, Liopyris K, Dusza SW, et al; International Skin Imaging Collaboration. Computer algorithms show potential for improving dermatologists' accuracy to diagnose cutaneous melanoma: results of the International Skin Imaging Collaboration 2017. *J Am Acad Dermatol*. 2020;82(3):622-627. doi:10.1016/j.jaad.2019.07.016

102. Maron RC, Utikal JS, Hekler A, et al. Artificial intelligence and its effect on dermatologists' accuracy in dermoscopic melanoma image classification: web-based survey study. *J Med Internet Res*. 2020;22(9):e18091. doi:10.2196/18091

103. Menzies SW, Bischof L, Talbot H, et al. The performance of SolarScan: an automated dermoscopy image analysis instrument for the diagnosis of primary melanoma. *Arch Dermatol*. 2005;141(11):1388-1396. doi:10.1001/archderm.141.11.1388

104. Menzies SW, Emery J, Staples M, et al. Impact of dermoscopy and short-term sequential digital dermoscopy imaging for the management of pigmented lesions in primary care: a sequential intervention trial. *Br J Dermatol*. 2009;161(6):1270-1277. doi:10.1111/j.1365-2133.2009.09374.x

105. Menzies SW, Ingvar C, Crotty KA, McCarthy WH. Frequency and morphologic characteristics of invasive melanomas lacking specific surface microscopic features. *Arch Dermatol*. 1996;132(10):1178-1182. doi:10.1001/archderm.1996.03890340038007

106. Menzies SW, Kreusch J, Byth K, et al. Dermoscopic evaluation of amelanotic and hypomelanotic melanoma. *Arch Dermatol*. 2008;144(9):1120-1127. doi:10.1001/archderm.144.9.1120

107. Menzies SW, Moloney FJ, Byth K, et al. Dermoscopic evaluation of nodular melanoma. *JAMA Dermatol*. 2013;149(6):699-709. doi:10.1001/jamadermatol.2013.2466

108. Moffatt CRM, Green AC, Whiteman DC. Diagnostic accuracy in skin cancer clinics: the Australian experience. *Int J Dermatol*. 2006;45(6):656-660. doi:10.1111/j.1365-4632.2006.02772.x

109. Moreno-Ramirez D, Ferrandiz L, Galdeano R, Camacho FM. Teledermatoscopy as a triage system for pigmented lesions: a pilot study. *Clin Exp Dermatol*. 2006;31(1):13-18. doi:10.1111/j.1365-2230.2005.02000.x

110. Nachbar F, Stolz W, Merkle T, et al. The ABCD rule of dermoscopy. High prospective value in the diagnosis of doubtful melanocytic skin lesions. *J Am Acad Dermatol*. 1994;30(4):551-559. doi:10.1016/S0190-9622(94)70061-3

111. Nilles M, Boedeker RH, Schill WB. Surface microscopy of naevi and melanomas–clues to melanoma. *Br J Dermatol*. 1994;130(3):349-355. doi:10.1111/j.1365-2133.1994.tb02932.x

112. Ojeda RM, Graells J. [Effectiveness of primary care physicians and dermatologists in the diagnosis of skin cancer: a comparative study in the same geographic area]. *Actas Dermosifiliogr*. 2011;102(1):48-52. doi:10.1016/j.ad.2010.06.020

113. Papageorgiou C, Apalla Z, Variaah G, et al. Accuracy of dermoscopic criteria for the differentiation between superficial basal cell carcinoma and Bowen's disease. *J Eur Acad Dermatol Venereol*. 2018;32(11):1914-1919. doi:10.1111/jdv.14995

114. Piccolo D, Crisman G, Schoinas S, Altamura D, Peris K. Computer-automated ABCD versus dermatologists with different degrees of experience in dermoscopy. *Eur J Dermatol*. 2014;24(4):477-481. doi:10.1684/ejd.2014.2320

115. Piccolo D, Ferrari A, Peris K, Diadone R, Ruggeri B, Chimenti S. Dermoscopic diagnosis by a trained clinician vs. a clinician with minimal dermoscopy training vs. computer-aided diagnosis of 341 pigmented skin lesions: a comparative study. *Br J Dermatol*. 2002;147(3):481-486. doi:10.1046/j.1365-2133.2002.04978.x

116. Piccolo D, Soyer HP, Chimenti S, et al. Diagnosis and categorization of acral melanocytic lesions using teledermoscopy. *J Telemed Telecare*. 2004;10(6):346-350. doi:10.1258/1357633042602017

117. Polesie S, Jergéus E, Gillstedt M, et al. Can dermoscopy be used to predict if a melanoma is in situ or invasive? *Dermatol Pract Concept*. 2021;11(3):e2021079. doi:10.5826/dpc.1103a79

118. Rampen FHJ, Casparie-van Velsen JI, van Huystee BEWL, Kiemeney LALM, Schouten LJ. False-negative findings in skin cancer and melanoma screening. *J Am Acad Dermatol*. 1995;33(1):59-63. doi:10.1016/0190-9622(95)90011-X

119. Rao BK, Marghoob AA, Stolz W, et al. Can early malignant melanoma be differentiated from atypical melanocytic nevi by in vivo techniques? part I, clinical and dermoscopic characteristics. *Skin Res Technol*. 1997;3(1):8-14. doi:10.1111/j.1600-0846.1997.tb00153.x

120. Rogers T, Marino ML, Dusza SW, et al. A clinical aid for detecting skin cancer: the Triage Amalgamated Dermoscopic Algorithm (TADA). *J Am Board Fam Med*. 2016;29(6):694-701. doi:10.3122/jabfm.2016.06.160079

121. Rolfe HM. Accuracy in skin cancer diagnosis: a retrospective study of an Australian public hospital dermatology department. *Australas J Dermatol*. 2012;53(2):112-117. doi:10.1111/j.1440-0960.2011.00855.x

122. Rosendahl C, Tschandl P, Cameron A, Kittler H. Diagnostic accuracy of dermatoscopy for melanocytic and nonmelanocytic pigmented lesions. *J Am Acad Dermatol*. 2011;64(6):1068-1073. doi:10.1016/j.jaad.2010.03.039

123. Rubegni P, Cevenini G, Nami N, et al. Dermoscopy and digital dermoscopy analysis of palmoplantar 'equivocal' pigmented skin lesions in caucasians. *Dermatology*. 2012;225(3):248-255. doi:10.1159/000343928

124. Rubegni P, Tognetti L, Argenziano G, et al. A risk scoring system for the differentiation between melanoma with regression and regressing nevi. *J Dermatol Sci*. 2016;83(2):138-144. doi:10.1016/j.jdermsci.2016.04.012

125. Russo T, Pampena R, Piccolo V, et al. The prevalent dermoscopic criterion to distinguish between benign and suspicious pink tumours. *J Eur Acad Dermatol Venereol*. 2019;33(10):1886-1891. doi:10.1111/jdv.15707

126. Seidenari S, Grana C, Pellacani G. Colour clusters for computer diagnosis of melanocytic lesions. *Dermatology*. 2007;214(2):137-143. doi:10.1159/000098573

127. Seidenari S, Pellacani G, Pepe P. Digital videomicroscopy improves diagnostic accuracy for melanoma. *J Am Acad Dermatol*. 1998;39(2 Pt 1):175-181. doi:10.1016/S0190-9622(98)70070-2

128. Seyed Ahadi M, Firooz A, Rahimi H, Jafari M, Tehranchinia Z. Clinical diagnosis has a high negative predictive value in evaluation of malignant skin lesions. *Dermatol Res Pract*. 2021;2021:6618990. doi:10.1155/2021/6618990

129. Skvara H, Teban L, Fiebiger M, Binder M, Kittler H. Limitations of dermoscopy in the recognition of melanoma. *Arch Dermatol*. 2005;141(2):155-160. doi:10.1001/archderm.141.2.155

130. Soyer HP, Argenziano G, Zalaudek I, et al. Three-point checklist of dermoscopy. A new screening method for early detection of melanoma. *Dermatology*. 2004;208(1):27-31. doi:10.1159/000075042

131. Stanganelli I, Serafini M, Bucch L. A cancer-registry-assisted evaluation of the accuracy of digital epiluminescence microscopy associated with clinical examination of pigmented skin lesions. *Dermatology*. 2000;200(1):11-16. doi:10.1159/000018308

132. Tognetti L, Cevenini G, Moscarella E, et al. An integrated clinical-dermoscopic risk scoring system for the differentiation between early melanoma and atypical nevi: the iDScore. *J Eur Acad Dermatol Venereol*. 2018;32(12):2162-2170. doi:10.1111/jdv.15106

133. Ulrich M, von Braunmuehl T, Kurzen H, et al. The sensitivity and specificity of optical coherence tomography for the assisted diagnosis of nonpigmented basal cell carcinoma: an

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

observational study. *Br J Dermatol*. 2015;173(2):428-435. doi:10.1111/bjd.13853

134. Unlu E, Akay BN, Erdem C. Comparison of dermatoscopic diagnostic algorithms based on calculation: the ABCD rule of dermatoscopy, the seven-point checklist, the three-point checklist and the CASH algorithm in dermatoscopic evaluation of melanocytic lesions. *J Dermatol*. 2014;41(7):598-603. doi:10.1111/1346-8138.12491

135. Witkowski AM, Łudzik J, DeCarvalho N, et al. Non-invasive diagnosis of pink basal cell carcinoma: how much can we rely on dermoscopy and reflectance confocal microscopy? *Skin Res Technol*. 2016;22(2):230-237. doi:10.1111/srt.12254

136. Yélamos O, Manubens E, Jain M, et al. Improvement of diagnostic confidence and management of equivocal skin lesions by integration of reflectance confocal microscopy in daily practice: prospective study in 2 referral skin cancer centers. *J Am Acad Dermatol*. 2020;83(4):1057-1063. doi:10.1016/j.jaad.2019.05.101

137. Youl PH, Baade PD, Janda M, Del Mar CB, Whiteman DC, Aitken JF. Diagnosing skin cancer in primary care: how do mainstream general practitioners compare with primary care skin cancer clinic doctors? *Med J Aust*. 2007;187(4):215-220. doi:10.5694/j.1326-5377.2007.tb01202.x

138. Yu C, Yang S, Kim W, et al. Acral melanoma detection using a convolutional neural network for dermoscopy images. *PLoS One*. 2018;13(3):e0193321. doi:10.1371/journal.pone.0193321

139. Zhu CY, Wang YK, Chen HP, et al. A deep learning based framework for diagnosing multiple skin diseases in a clinical environment. *Front Med (Lausanne)*. 2021;8:626369. doi:10.3389/fmed.2021.626369

140. Zortea M, Schopf TR, Thon K, et al. Performance of a dermoscopy-based computer vision system for the diagnosis of pigmented skin lesions compared with visual evaluation by experienced dermatologists. *Artif Intell Med*. 2014;60(1):13-26. doi:10.1016/j.artmed.2013.11.006

141. Warshaw EM, Gravely AA, Nelson DB. Accuracy of teledermatology/teledermoscopy and clinic-based dermatology for specific categories of skin neoplasms. *J Am Acad Dermatol*. 2010;63(2):348-352. doi:10.1016/j.jaad.2009.10.037

142. Rajpara SM, Botello AP, Townend J, Ormerod AD. Systematic review of dermoscopy and digital dermoscopy/ artificial intelligence for the diagnosis of melanoma. *Br J Dermatol*. 2009;161(3):591-604. doi:10.1111/j.1365-2133.2009.09093.x

143. Hao T, Meng XF, Li CX. A meta-analysis comparing confocal microscopy and dermoscopy in diagnostic accuracy of lentigo maligna. *Skin Res Technol*. 2020;26(4):494-502. doi:10.1111/srt.12821

144. Carapeba MOL, Alves Pineze M, Nai GA. Is dermoscopy a good tool for the diagnosis of lentigo maligna and lentigo maligna melanoma? A meta-analysis. *Clin Cosmet Investig Dermatol*. 2019;12:403-414. doi:10.2147/CCID.S208717

145. Lan J, Wen J, Cao S, et al. The diagnostic accuracy of dermoscopy and reflectance confocal microscopy for amelanotic/hypomelanotic melanoma: a systematic review and meta-analysis. *Br J Dermatol*. 2020;183(2):210-219. doi:10.1111/bjd.18722

146. Nelson KC, Swetter SM, Saboda K, Chen SC, Curiel-Lewandrowski C. Evaluation of the number-needed-to-biopsy metric for the diagnosis of cutaneous melanoma: a systematic review and meta-analysis. *JAMA Dermatol*. 2019;155(10):1167-1174. doi:10.1001/jamadermatol.2019.1514

147. Fried LJ, Tan A, Berry EG, et al. Dermoscopy proficiency expectations for US dermatology resident physicians: results of a modified Delphi survey of pigmented lesion experts. *JAMA Dermatol*. 2021;157(2):189-197. doi:10.1001/jamadermatol.2020.5213

148. Tran T, Cyr PR, Verdieck A, et al. Expert consensus statement on proficiency standards for dermoscopy education in primary care. *J Am Board Fam Med*. 2023;36(1):25-38. doi:10.3122/jabfm.2022.220143R1

149. Mangione CM, Barry MJ, Nicholson WK, et al; US Preventive Services Task Force. Screening for skin cancer: US Preventive Services Task Force Recommendation statement. *JAMA*. 2023;329(15):1290-1295. doi:10.1001/jama.2023.4342

150. Oliveria SA, Heneghan MK, Cushman LF, Ughetta EA, Halpern AC. Skin cancer screening by dermatologists, family practitioners, and internists: barriers and facilitating factors. *Arch Dermatol*. 2011;147(1):39-44. doi:10.1001/archdermatol.2010.414

151. Shellenberger RA, Tawagi K, Kakaraparthi S, Albright J, Nabhan M, Geller AC. Are primary care residents trained to perform skin cancer examinations? *J Gen Intern Med*. 2018;33(11):1839-1841. doi:10.1007/s11606-018-4572-x

152. Fleischer AB Jr, Herbert CR, Feldman SR, O'Brien F. Diagnosis of skin disease by nondermatologists. *Am J Manag Care*. 2000;6(10):1149-1156.

153. Glazer AM, Rigel DS. Analysis of trends in geographic distribution of US dermatology workforce density. *JAMA Dermatol*. 2017;153(5):472-473. doi:10.1001/jamadermatol.2016.6032

154. Nguyen C, Nguyen QB, Lauck K, Hebert A. Five years later: continuing disparity in the geographic density and distribution of US dermatologists. *Skin (Milwood, NY)*. 2023;7(3):811-816. doi:10.25251/skin.7.3.9

155. Vaidya T, Zubritsky L, Alikhan A, Housholder A. Socioeconomic and geographic barriers to dermatology care in urban and rural US populations. *J Am Acad Dermatol*. 2018;78(2):406-408. doi:10.1016/j.jaad.2017.07.050

156. Young AT, Xiong M, Pfau J, Keiser MJ, Wei ML. Artificial intelligence in dermatology: a primer. *J Invest Dermatol*. 2020;140(8):1504-1512. doi:10.1016/j.jid.2020.02.026

157. Schneider JS, Moore DH II, Mendelsohn ML. Screening program reduced melanoma mortality at the Lawrence Livermore National Laboratory, 1984 to 1996. *J Am Acad Dermatol*. 2008;58(5):741-749. doi:10.1016/j.jaad.2007.10.648

158. Kashani-Sabet M, Leachman SA, Stein JA, et al. Early detection and prognostic assessment of cutaneous melanoma: consensus on optimal practice and the role of gene expression profile testing. *JAMA Dermatol*. 2023;159(5):545-553. doi:10.1001/jamadermatol.2023.0127

159. Johnson MM, Leachman SA, Aspinwall LG, et al. Skin cancer screening: recommendations for data-driven screening guidelines and a review of the US Preventive Services Task Force controversy. *Melanoma Manag*. 2017;4(1):13-37. doi:10.2217/mmt-2016-0022

160. Whiting G, Stocks N, Morgan S, et al. General practice registrars' use of dermoscopy: prevalence, associations and influence on diagnosis and confidence. *Aust J Gen Pract*. 2019;48(8):547-553. doi:10.31128/AJGP-11-18-4773

161. Williams NM, Marghoob AA, Seiverling E, Usatine R, Tsang D, Jaimes N. Perspectives on dermoscopy in the primary care setting. *J Am Board Fam Med*. 2020;33(6):1022-1024. doi:10.3122/jabfm.2020.06.200238

© 2024 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Copyright of JAMA Dermatology is the property of American Medical Association and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.